Introduction

Contemporary Digital Electronic Systems — Chapter 1

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Why This Chapter Matters

Your phone, the car it rode in, the appliance it is charging on: each has a digital electronic system at its core. This chapter sets up the vocabulary and mental model for the rest of the module.

After this chapter, you can:

1. Distinguish analog from digital quantities.
2. Describe the voltage-range abstraction behind logic values.
3. Qualify claims such as “repeatable” and “noise-tolerant”.
4. Name the digital IC types—ASIC, FPGA, CPLD, MCU, MPU, SoC.
5. Read a timing diagram.

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An Opening Question

A thermometer, a speedometer and a stopwatch all measure things.

Which should use a digital representation internally, and which an analog one?

By the end of this chapter you should be able to defend an answer.

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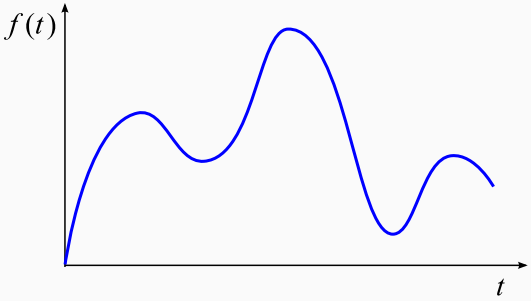
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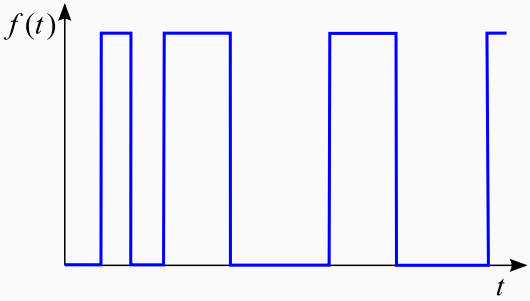
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Analog versus Digital



Analog: any value in a range



Digital: a few discrete values

- Digital circuits use only two logic values: TRUE/FALSE, ON/OFF, HIGH/LOW.
- We therefore use binary numbers; each digit is a bit (*bi*nary digi*t*).

Why Digital?

Advantages

- Tractable design: Boolean algebra, not differential equations
- Repeatable behaviour within specified inputs and timing
- Noise tolerant: a voltage band maps to one symbol
- Durable storage within rated conditions
- Programmable: software or FPGA
- Rich processing: compression, encryption, filtering
- Controllable precision: more bits, more fidelity

Trade-offs

- Needs an ADC/DAC at the analog interface
- Quantization and sampling error
- Analog can be smaller, cheaper or lower power for narrow tasks
- Sampling and processing add latency

These claims hold *within specified operating conditions*, not as absolute properties of hardware.

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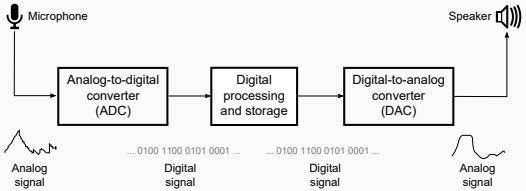
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Where Digital Processing Dominates

Domain	Digital pipeline
Audio	ADC, filtering, noise suppression, codec (AAC, Opus, LC3), transport, decode, DAC
Video	Sensor readout, image signal processing, H.264 / HEVC / AV1 encoding in hardware blocks
Imaging	CMOS sensor, ISP, JPEG or video codec
Sensing and control	MCU with built-in ADC; filtering, calibration, decisions and communication in software

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A Digital System for Analog Signals



- ADC converts the analog signal; all processing is then digital; a DAC restores analog before amplification.
- Sampling is periodic; each sample maps to the nearest code (here 00, 01, 10, 11).

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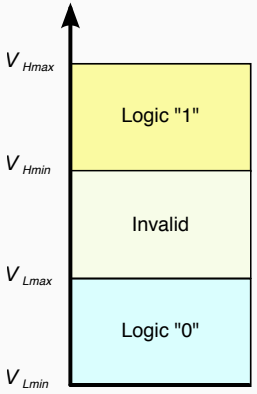
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Logic Levels Are Voltage Ranges

Notation	Definition
V_{DD} or V_{CC}	Supply voltage
V_{Hmin}	Minimum voltage for HIGH
V_{Lmax}	Maximum voltage for LOW
0 V	Ground

- Between the two ranges is undefined; avoid it in steady state.



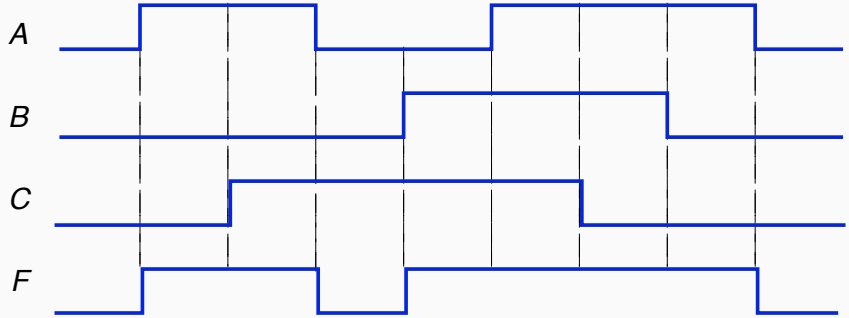
TTL, CMOS and Supply Voltage

- TTL: older, 5 V supply (V_{CC}).
- CMOS: almost all modern logic, supply V_{DD}.
- CMOS supplies keep falling: 3.3, 2.5, 1.8, 1.2, 1.05 V and below.
- Reason: dynamic power scales as CV^2f.
- Speed gains come from process scaling, not lower voltage alone.

Illustrative 3.3 V CMOS convention

LOW: 0 to $V_{DD}/3$ ($\approx$ 1.1 V); HIGH: $2V_{DD}/3$ ($\approx$ 2.2 V) to V_{DD}.
So 1 V is LOW, 2 V is undefined, 3 V is HIGH. Real parts specify $V_{IL}, V_{IH}, V_{OL}, V_{OH}$ in the datasheet.

Timing Diagrams



- A timing diagram plots logic values against time, for one or more signals.
- HIGH and LOW are drawn as straight lines; transitions as vertical lines.
- Real transitions take time, but are short compared with the time between them.

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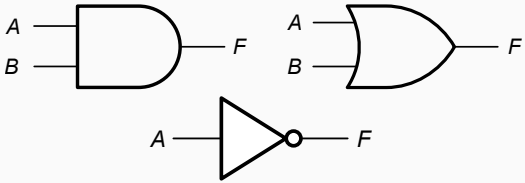
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- AND: HIGH only if both inputs HIGH.
- OR: HIGH if at least one input HIGH.
- NOT: inverts its input.
- Output depends on present input only: combinational.

Flip-flops

Retain a 0 or 1. The stored value is the state. Output depends on past inputs too: sequential circuits have *memory*.

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Levels of Digital Circuits

Level	Examples
System	Processors, memories
Module	Multiplexers, decoders, counters, registers
Primitive	Gates, flip-flops
Device	Transistors, resistors, capacitors

Lower levels are joined to build the level above.

- An IC puts many transistors on one silicon die.
- Dies are made many to a wafer, tested, cut, then packaged.
- Historical classes by transistor count are only approximate.

Class	Approx. transistors
SSI	Fewer than 100
MSI	Hundreds to thousands
LSI	Thousands to tens of thousands+
VLSI	Hundreds of thousands to millions+
ULSI	No accepted threshold

A modern SoC holds tens of billions of transistors, so a design-based classification is more useful.

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Modern IC Types

Type	Function determined by	Flexibility	Typical use
ASIC	Fabricated design	Fixed after fabrication	Application-specific products
FPGA	Configuration bitstream	Reconfigurable	Prototyping, specialized computing
CPLD	Device configuration	Reconfigurable	Glue logic, small control
MCU	Firmware	Programmable	Embedded control
MPU	Software	Programmable	General-purpose computing
SoC	Hardware design and software	Depends on implementation	Phones, tablets

Categories overlap: an SoC can be an ASIC; an MCU can be implemented as an ASIC.

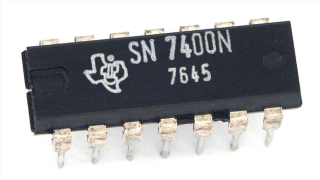
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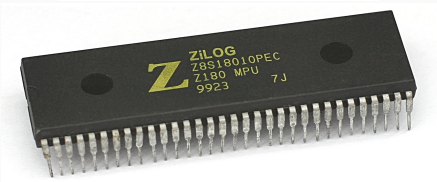
MCU, MPU and SoC

MPU	MCU	SoC
General-purpose processor; relies on external memory and peripherals.	Core, memory and peripherals (timers, communication, ADC) on one chip.	Most of a computing system on a chip: cores, memory controllers, GPU, accelerators.

Modern designs are described in an HDL (Verilog or VHDL) and handled by EDA tools—see Chapter 16.

Packaging





DIP: through-hole, 0.1 in pin pitch

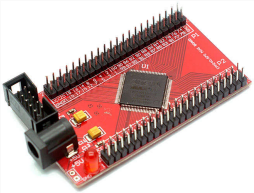


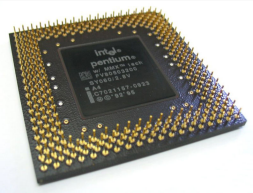
SMT: SOIC, PLCC, QFP

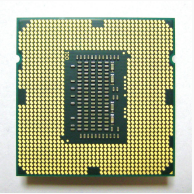
- Packages protect the die and expose pins.
- DIP: breadboard-friendly, but bulky.
- SMT packages dominate: smaller, suited to automated assembly.
- Written DIP*n*; *n* is the pin count.

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Packages: Then and Now







MAX II CPLD | Pentium (1996): 4.5 million transistors | i5-750 (2009): 774 million

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- A digital signal carries one of two logic values; each is a voltage range.
- “Repeatable”, “noise-tolerant” and “cheaper” hold only under stated conditions.
- CMOS has displaced TTL; lower V_{DD} cuts dynamic power ($\propto CV^2f$).
- Gates are combinational; flip-flops are sequential. Modules compose primitives; systems compose modules.
- Classify modern ICs as ASIC, FPGA, CPLD, MCU, MPU or SoC.

Looking ahead

Chapter 2 shows how patterns of bits represent integers, fractions, characters and sequences.